We would like to express our sincere gratitude to the reviewers for their valuable feedback and insightful comments on our manuscript. We appreciate the time and effort they have dedicated to evaluating our work, and we have carefully considered each of their suggestions. Below, we provide detailed responses to the points raised, along with explanations of the revisions made to address them. For easier readability, we have numbered each of the comments, and responses are provided in purple.

Reviewer: 1

1. Page 4, Space 7: Mary Lea Heger was indeed the first to notice some diffuse unidentified bands in the spectra of B-stars in 1922. However, their interstellar origin was first established by Paul Merrill years later in a paper in 1936: Merrill, P. 1936, AJ, 83, 126. This attribution should be corrected.

   We have corrected the attribution. The revised text now states: "The DIBs were first recorded by Heger[7], with their interstellar origin later established by Merrill[8]."

2. Page 4, Space 23: 30 Å may no longer be the largest FWHM among DIBs. Maíz Apellániz et al. (2021, MNRAS, 501, 2487) reported the discovery of λ7700 DIB with a FWHM of 177 Å. In astronomical spectra taken from the ground, this DIB is largely obscured by the telluric A-band and has, therefore, escaped detection in groundbased spectroscopic surveys. The interstellar appearance of this new band appears to be like that of typical σ-type DIB.

   Thank you for pointing this out. We have now included this in our introduction: "The DIB widths (measured as the full width at half maximum, FHWM) vary from 0.5 to 30 Å with a recent discovery of a new DIB at 7700 Å with FWHM of 177 Å[25]. However, the majority of DIBs have a width between 0.5 and 3 Å[5]".

3. Page 6, Space 3: Note, Lai et al. (2020, MNRAS, 492, 5853) presented data concerning substantial variability of the FWHM of the λ4428 DIB, which were also attributed to variations in the rotational temperature in the different ISM environments hosting the carrier in question. Just another example ...

   We have now added a sentence at the end of fourth paragraph of our introduction to include example of papers that suggest that variations are caused by variations in rotational temperature. It has been proposed that change in rotational temperature causes variations in DIB profiles other than λ6614 as well, such as $\lambda\lambda$5797, 6379[22] and λ4428[36].

4. Page 6, Space 37: I was troubled by the fact that the work of MacIsaac et al. (2022), based on the identical astronomical data set as that used in the present publication and published with largely the same set of co-authors as here, concluded that the likely rotational constant B of the λ6614 DIB carrier is larger by about a factor of 10 compared to the value arrived at in the present investigation. This discrepancy needs to be at least commented upon, if not explained.

   Thank you for pointing this out; our original text may not have been sufficiently clear. We have now revised the first paragraph of the discussion section to state: "This is significantly smaller than what Cami et al. [20] and MacIsaac et al. [22] found which was derived based exclusively on the peak separation and not contour simulations. Moreover, they assume that the potential carrier is a linear or spherical species".

5. Page 8, Space 36: I suggest, replacing “often” with “almost always”. Under interstellar conditions, the time scales for electronic and vibrational de-excitations are orders of magnitude shorter than the time scales between successive photon absorptions from the highly diluted interstellar radiation field, making it rather safe to assume that the systems are in the electronic and vibrational ground state, with a distribution of rotational states still present due to the very low Einstein A-values for rotational transitions, when a new optical photon is absorbed.

   We agree that "almost always" would be a better choice of word. Revised version: "The lower state will almost always be the ground state when considering interstellar absorption".

6. Page 15, Space 34: While a molecular size of 20 carbon atoms may indeed be the lower size limit of particles able to survive under typical conditions in the diffuse ISM (photon energy < 13.6 eV; see Iida et al. 2022, MNRAS, 514, 844), the surviving particles most likely are not PAHs. In such environments, PAHs are expected to typically lose their external H-atoms for NC 40 or less.

We would like to highlight that we consider oblate top geometry for this study and not coronene specifically. To clarify, the revised version states: If the DIB carriers are PAH-like planar molecules, their size should be at least about 20 carbon atoms since small PAHs are thought to be unstable against photo-dissociation in the ISM[41]. That is about the size of coronene ($C_{24}H_{12}$), ~~the smallest PAH with the oblate top geometry that we focus on in this paper~~. Coronene has $B = 1.11 \times 10^{-2}$ cm^{-1} [42], and we thus chose as upper limit a slightly higher value of $B = 5 \times 10^{-2}$ cm^{-1} corresponding to slightly smaller species.

7. Page 26, Figure 7: If I read the colors correctly, the authors compare models with one another and observed profiles with one another in each of the three panels. Would it not make better sense to compare each observed profile with its corresponding model? The legends and the labels on the axes are so small, that they will be indecipherable when reduced to the actual publication format. They should be presented in larger-size fonts.

   Thank you for your comment on this figure (Note that due to the changes in previous figures, this is now figure 2). It is an important figure for the paper and thus we would like to make it as easy to comprehend as possible. We have now increased the font size of the legends, labels and the x and y ticks sizes. Moreover, instead of having the first two figures side-by-side, we have all three figures in a column to allow more room per figure. We considered your suggestion about comparing each observed profile with its corresponding model, however, we think that comparing models with one another and observed profiles with one another is a good idea because we would like to highlight the differences between observed profiles with narrowest and widest peak separation. Moreover, placing models on top of each other highlights better how changing a certain parameter would change the model in the context of λ6614 DIB.

8. Page 34, Space 3: The discovery of a positive correlation between the line widths deduced from the 6614 DIB profile fittings and the corresponding CH+ line widths is potentially the most important, lasting result from the present study. It would really be helpful to see a graph of these data to illustrate this correlation rather than just quoting the Pearson correlation coefficient.

   We agree that it is a good idea to add such a figure. We have now added a figure (Figure 7) in the discussion section that shows the linewidth of λ6614 DIB vs. linewidth of CH^+.

9. Page 34, Space 8: The potential association of the λ6614 DIB carrier molecule with environments characterized by the presence of CH+ should be further explored and substantiated. The first step would be to plot the EW(6614) values against the EW(CH+) values. Naturally, most likely there will be a positive correlation, as both values are correlated with the increasing column densities of interstellar matter, given that the sampled stars have a range of $0.21 < E(B\text{-}V) < 0.56$. However, suppose the λ6614 DIB carrier and CH+ prefer the same limited environments. In that case, the correlation between their equivalent widths should be significantly tighter than the correlations between EW(6614) and E(B-V) or between EW(6614) and N(H). If this were found to be the case, the currently rather vague suggestion of a physical association would be supported more solidly. If not, the connection between these two different carriers is more likely a rather tenuous one.

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Reviewer: 2

10. There are two papers by Bernstein et al. 2015 and 2018 that are on the general subject of spectral modeling of the 6614 DIB and should be briefly mentioned in the introduction (full ref. given at end of comments). The model assumptions in each paper are different from the paper under review, but good fits are obtained, which further demonstrate the non-uniqueness of modeling the 6614 DIB spectral profile.

    Thank you for this suggestion. We have now included references to Bernstein et al. 2015 and 2018 in the fourth paragraph of the introduction: "In an alternative scenario, Bernstein et al. [34] reproduced the λ6614 DIB by modelling superposition of two parallel transitions of a molecule with prolate top symmetry. Furthermore, Bernstein et al. [35] performed spectral deconvolution to find that the λ6614DIB can be reproduced as a combination of doublet and a redtail component arising from different carriers."

11. Fig. 1 is unnecessary. A sentence or two on the validated against PGOPHER is sufficient.

    We have removed Fig. 1 as recommended.

12. Figures 2-6 nicely summarize the parameter variations. I recommend that for each figure, all the related curves be placed in a single plot and a summary figure be made with the five adjacent plots forming a panel figure. It would also be helpful to include one representative data profile in each of the five component plots. This makes it easier for a reader to compare the effects of the parameter variations. Given this panel figure the huge number of tiny plots in A and B are not necessary. The authors can mention them in the text and direct the reader to one of their web sites to be viewed or downloaded

    Thank you for this suggestion. We think combining Figures 2-6 in a single plot is a great idea as the reader can compare the effect of all parameter variations on one page. We have replaced Figures 2-6 with a new Figure 1, which now comprises five component plots, one for each parameter. We also included a representative data profile in each of the five component plots. Plots from Appendix A and B are removed now.

13. I found sections 4.3, 5, and 6 confusing rather than clarifying. Too much information and too many plots to digest. The preceding sections nicely describe the model and its parameters and their range; so, at this point it would make sense to present the fitting results. Many of the points made in sections 4.3-6 can be incorporated in the fitting or perhaps the discussion sections and results. For instance the figures showing the fits also contain all the observed profiles, so the discussion of profile variations can be based on these figures.

    Thank you for pointing out that sections 4.3, 5, and 6 may be overly detailed and potentially confusing. After considering your suggestion, we agree that sections 4.3 and 5 are not essential to the main flow of the paper and have removed them. However, we believe section 6 is important as it emphasizes that variations in the λ6614 profile cannot be solely attributed to changes in rotational temperature, as has been previously assumed. We have retained section 6 (which is now section 5) to highlight this key finding.

14. Combine figs. 8 & 49 as side by side plots and add the independent sightline fits for each observation. This allows for removal of figs, 37-49 from the appendix. This new fig. replaces current Fig.7 and supports much of the discussion in sections 4.3-6. For clarity, the fit summary figure should include two adjacent panels, one for the EDIBLES fits and one for the Galazutinov fits, and the fit for each source should include the data and the two dependent and independent sightline fits. This results in a series of displaced curves, with three per source.

    We have included several figures in one plot now. We have combined Figs. 8 and 49 into a single side-by-side plot, which is now Figure 5, and included the independent sightline fits for each source, showing three profiles per source. This allowed us to remove Figs. 37-49 from the appendix. However, we retained Figure 7 as a separate figure, as it highlights the cause of variations in the observed profiles that may not be immediately evident in the new Figure 5.

15. The authors should consider the possibility of two overlapping PQR rotational profiles. There are several arguments supporting this idea. First, consider the shift in the band center of a 13C substituted large PAH. According to eq.(7) in Webster (2004) the shift in a VIS band center for a single 13C substituted in a PAH is approximately v 18/Nc, where Nc is the number of C atoms in the PAH. Assuming, for example, Nc=50, v 0.36 cm-1. Even assuming a larger shift, say a factor of two, this still implies that the isotopic shift should

reside well within the observed profile width of 2 cm-1. For clarity, Wester's model attributes each observed feature to a different 13C isotope, whereas I attribute three adjacent features to the PQR branches of a single 13C isotope.

Second, spectral deconvolution (see fig. 5 in Bernstein et al. 2018) demonstrates that the observed feature is likely composed of two, nearly identical and slightly shifted PQR rotational profiles. Third, profile fitting of two high spectral resolution profiles from different sources results in improved fits (figs 7 and 8 in Bernstein et al. 2018). Both fits are consistent with a profile shift of 0.33 +/- 0.03 cm-1 (this uncertainty is within the resolution of 0.075 cm-1) and relative profile heights of 0.67 (the pure 12C component) and 0.33 (13C containing component). Assuming an average interstellar 12C/13C ratio of R=76 (1.31 12C abundance of the profile carrier is around Nc 30 (i.e., (1-0.0131)30=0.67. The 13C abundance may vary for different sources, so this parameter should be allowed to vary. This is an independent determination of Nc and suggests a PAH comparable to ovalene, C34H14, with B 0.054 cm-1. Most of the spectral fits in the current work, while plausible, show significant errors in the regions of the PQR peaks, which should be significantly improved by a two-profile model.

Thank you for your suggestion; we have given it careful consideration. We acknowledge that there are some discrepancies in the fits around the PQR peaks in our results. Adding the ^{13}C abundance as a free parameter could potentially help reduce these errors. However, we believe this is better suited for future work. In this study, our primary aim is to reproduce profile variations across 12 different sightlines, rather than achieving the best possible fit for a single profile.

16. Previous analysis (Bernstein et al. 2018 ApJ figs 7&8) also shows that the red wing feature overlaps the blue side of the main feature. They show that the main feature fitting should not extend below -1 cm-1, which is equivalent to the 95% criteria adopted by the authors

    Not sure if any changes to the paper are requested, how to respond ?

17. Assuming some 550 DIBS spanning the vis range, 400-700 nm, the average spacing between DIBS is 20 cm-1. Assuming random band positions within each sub-interval, the number of bands located within 2 cm-1 of another band is of order 55o(2/20) 56. Thus, it is entirely possible that the red wing is a DIB from a different molecule.

    Not sure if any changes to the paper are requested, how to respond ?

18. Note that the Bernstein et al. 2015 and 2018 papers recognized anomalous broadening for the 6614 DIB and discuss possible sources. They also point out that the highly correlated 6196 and 6614 DIBS show the same anomalous broadening (Bernstein et al. 2018).

    Not sure if any changes to the paper are requested, how to respond ?

19. I am not familiar with origin and transient nature of CH+. Can you add a little detail, with an estimate of its transient lifetime. It might be interesting to speculate on the potential transient lifetime of the 6614 DIB which might be due to UV photodissociation?

    Formation of CH^+ is via this endothermic reaction: $C^+ + H_2 \rightarrow CH^+ + H$. Once formed, it can be easily destructed by collisions with H, H_2 or e^-. Formation and destruction rates of CH^+ are given by Godard et al. (2023) which are temperature dependent. We are not sure if it is straightforward to speculate the transient lifetime of λ6614 DIB carrier based on transient lifetime of CH^+, however, we think a future project dedicated to analyzing the potential connection between CH^+ and λ6614 DIB carrier would be worthwhile.

20. One of the key study results, the very small spread in Trot, seems odd to me. Is there some measure of Tkin in the single molecular cloud sources you studied? It would be interesting if the spread in Tkin were also small.

    Table 1 indicates the T_{kin} measurements for some of our target sightlines from Fan et al. (2024) which were obtained based on C_2 profiles:

    However, we are not sure if the rotational temperature of the λ6614 carrier would depend on the kinetic temperature.

Sightline	$\mathbf{T_{kin}}$
HD 23180	$37.7^{+2.3}_{-2.5}$
HD 24398	$51.1^{+5.8}_{-3.5}$
HD 144470	-
HD 147683	-
HD 147683	$26.4^{+3.6}_{-2.1}$
HD 149757	$42.6^{+1.1}_{-1.9}$
HD 166937	-
HD 170740	$33.2^{+2.8}_{-1.9}$
HD 184915	-
HD 185418	$31.6^{+5.3}_{-4.8}$
HD 185859	-
HD 203532	$39.5^{+1.0}_{-1.7}$

Table 1: Table of Sightline and Kinetic Temperature with Uncertainties

References

Fan, H., Rocha, C. M. R., Cordiner, M., et al. 2024, Astron. Astrophys., 681, A6

Godard, B., Pineau des Forêts, G., Hennebelle, P., Bellomi, E., & Valdivia, V. 2023, Astron. Astrophys., 669, A74